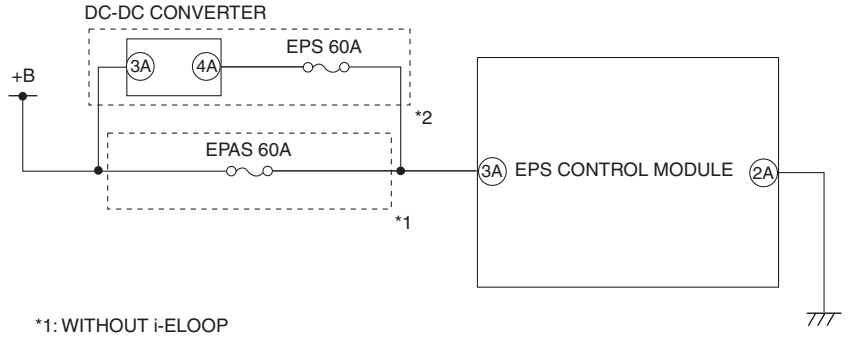
DTC U3003:16 [EPS CONTROL MODULE]

id060230812800

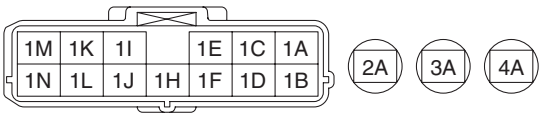
DTC	U3003:16	Battery power supply
DETECTION CONDITION		• Voltage of 8 V or less detected by EPS control module internal battery voltage observation
FAIL-SAFE FUNCTION		• Power steering malfunction indicator light Illuminates • EPS motor current control is maintained by gradually decreasing the motor control current. However, control is inhibited if the power supply voltage is the specified value of less
POSSIBLE CAUSE		**Without i-ELOOP** • DTC is stored in PCM • Battery malfunction • Generator malfunction • Fuse (EPAS 60A) malfunction • Open or short to GND circuit in wiring harness between battery and EPS control module terminal 3A • Open circuit in wiring harness between EPS control module terminal 2A and body ground • Poor connection at each connector **With i-ELOOP** • DTC is stored in PCM • Battery malfunction • DTC is stored in DC-DC converter (i-ELOOP) • Fuse (EPS 60A) malfunction • Open or short to GND circuit in wiring harness between DC-DC converter terminal 4A and EPS control module terminal 3A • Open circuit in wiring harness between EPS control module terminal 2A and body ground • Poor connection at each connector

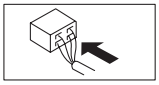


*1: WITHOUT i-ELOOP

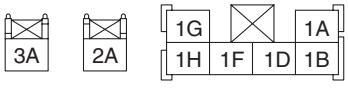
*2: WITH i-ELOOP

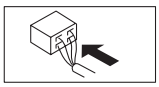
DC-DC CONVERTER
WIRING HARNESS-SIDE CONNECTOR





EPS CONTROL MODULE
WIRING HARNESS-SIDE CONNECTOR





Diagnostic Procedure
Without i-ELOOP

Step	Inspection		Action
1	**VERIFY PCM DTC** • Using the M-MDS, perform the PCM DTC inspection. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Are any DTCs present?	Yes	Go to applicable DTC inspection. (See DTC TABLE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See DTC TABLE [SKYACTIV-D 1.5].) DTC troubleshooting completed, then go to the next step.
		No	Go to the next step.

Step	Inspection		Action
2	**INSPECT BATTERY** • Inspect the battery. (See BATTERY INSPECTION.) • Is there any malfunction?	Yes	Charge or replace the battery, then go to Step7. (See BATTERY RECHARGING.) (See BATTERY REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See BATTERY REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.
3	**INSPECT GENERATOR** • Inspect the generator. (See GENERATOR INSPECTION [WITHOUT i-ELOOP (SKYACTIV-G 1.3, SKYACTIV-G 1.5)].) (See GENERATOR INSPECTION [WITHOUT i-ELOOP (SKYACTIV-D 1.5)].) • Is there any malfunction?	Yes	Replace the generator, then go to Step7. (See GENERATOR REMOVAL/INSTALLATION [WITHOUT i-ELOOP (SKYACTIV-G 1.3, SKYACTIV-G 1.5)].) (See GENERATOR REMOVAL/INSTALLATION [WITHOUT i-ELOOP (SKYACTIV-D 1.5)].)
		No	Go to the next step.
4	**INSPECT FUSE CONDITION** • Is the fuse (EPAS 60A) normal?	Yes	Go to the next step.
		No	• If the fuse is blown: — Refer to the wiring diagram and verify whether or not there is a common connector between fuse and EPS control module terminal 3A. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. • Replace the fuse. • If the fuse is damaged: — Replace the fuse. Go to Step7.
5	**INSPECT EPS CONTROL MODULE POWER SUPPLY CIRCUIT FOR OPEN CIRCUIT** • Switch the ignition OFF. • Disconnect the EPS control module connector. • Switch the ignition ON (engine off). • Measure the voltage between EPS control module terminal 3A (wiring harness-side) and body ground. • Is the voltage **10 V or more**?	Yes	Go to the next Step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between battery and EPS control module terminal 3A. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step7.

Step	Inspection		Action
6	**INSPECT EPS CONTROL MODULE GROUND CIRCUIT FOR OPEN CIRCUIT OR POOR GROUND** • Switch the ignition OFF. • Disconnect the EPS control module connector. • Inspect for continuity between EPS control module terminal 2A (wiring harness-side) and body ground. • Is the resistance **0—1 ohm**?	Yes	Go to the next step.
		No	• If the open circuit could be detected in the wiring harness: — Refer to the wiring diagram and verify whether or not there is a common connector between EPS control module terminal 2A and body ground. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. — Go to the next step. • If the resistance is out of specification: — Refer to the wiring diagram and verify whether or not there is a common connector between EPS control module terminal 2A and body ground. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a poor ground. — Go to the next step.
7	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Using the M-MDS, clear the DTC from the EPS control module. (See CLEARING DTC [EPS CONTROL MODULE].) • Using the M-MDS, perform the EPS control module DTC inspection. (See DTC INSPECTION [EPS CONTROL MODULE].) • Is the same DTC present?	Yes	Repeat the inspection from Step1. If the malfunction recurs, replace the steering column, then go to the next step. (See STEERING WHEEL AND COLUMN REMOVAL/INSTALLATION.)
		No	Go to the next step.
8	**VERIFY NO DTC IS PRESENT** • Are any DTCs present?	Yes	Go to applicable DTC inspection. (See DTC TABLE [EPS CONTROL MODULE].)
		No	DTC troubleshooting completed.

With i-ELOOP

Step	Inspection		Action
1	**VERIFY PCM DTC** • Using the M-MDS, perform the PCM DTC inspection. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Are any DTCs present?	Yes	Go to applicable DTC inspection. (See DTC TABLE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See DTC TABLE [SKYACTIV-D 1.5].) DTC troubleshooting completed, then go to the next step.
		No	Go to the next step.
2	**INSPECT BATTERY** • Inspect the battery. (See BATTERY INSPECTION.) • Is there any malfunction?	Yes	Charge or replace the battery, then go to Step7. (See BATTERY RECHARGING.) (See BATTERY REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See BATTERY REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.

Step	Inspection		Action
3	**VERIFY DC-DC CONVERTER DTC** • Using the M-MDS, perform the DC-DC converter DTC inspection. (See DTC INSPECTION [i-ELOOP].) • Are any DTCs present?	Yes	Go to applicable DTC inspection. (See DTC TABLE [i-ELOOP].) DTC troubleshooting completed, then go to the next step.
		No	Go to the next step.
4	**INSPECT FUSE CONDITION** • Is the fuse (EPS 60A) normal?	Yes	Go to the next step.
		No	• If the fuse is blown: — Refer to the wiring diagram and verify whether or not there is a common connector between fuse and EPS control module terminal 3A. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. • Replace the fuse. • If the fuse is damaged: — Replace the fuse. Go to Step7.
5	**INSPECT EPS CONTROL MODULE POWER SUPPLY CIRCUIT FOR OPEN CIRCUIT** • Switch the ignition OFF. • Disconnect the EPS control module connector. • Switch the ignition ON (engine off). • Measure the voltage between EPS control module terminal 3A (wiring harness-side) and body ground. • Is the voltage **10 V or more**?	Yes	Go to the next Step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between DC-DC converter terminal 4A and EPS control module terminal 3A. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step7.

Step	Inspection		Action
6	**INSPECT EPS CONTROL MODULE GROUND CIRCUIT FOR OPEN CIRCUIT OR POOR GROUND** • Switch the ignition OFF. • Disconnect the EPS control module connector. • Inspect for continuity between EPS control module terminal 2A (wiring harness-side) and body ground. • Is the resistance **0—1 ohm**?	Yes	Go to the next step.
		No	• If the open circuit could be detected in the wiring harness: — Refer to the wiring diagram and verify whether or not there is a common connector between EPS control module terminal 2A and body ground. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. — Go to the next step. • If the resistance is out of specification: — Refer to the wiring diagram and verify whether or not there is a common connector between EPS control module terminal 2A and body ground. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a poor ground. — Go to the next step.
7	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Using the M-MDS, clear the DTC from the EPS control module. (See CLEARING DTC [EPS CONTROL MODULE].) • Using the M-MDS, perform the EPS control module DTC inspection. (See DTC INSPECTION [EPS CONTROL MODULE].) • Is the same DTC present?	Yes	Repeat the inspection from Step1. If the malfunction recurs, replace the steering column, then go to the next step. (See STEERING WHEEL AND COLUMN REMOVAL/ INSTALLATION.)
		No	Go to the next step.
8	**VERIFY NO DTC IS PRESENT** • Are any DTCs present?	Yes	Go to applicable DTC inspection. (See DTC TABLE [EPS CONTROL MODULE].)
		No	DTC troubleshooting completed.